

Assignment 0.1

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Declaration:

*"I have read, and I understand the plagiarism provisions in the General Regulations of the University Calendar for*

*the current year, found at <http://www.tcd.ie/calendar>.*

*I have also completed the Online Tutorial on avoiding plagiarism 'Ready Steady Write', located at <http://tcd-ie.libguides.com/plagiarism/ready-steady-write>."*

As part of an assignment for a previous module, I created an entity-relationship (ER) diagram for an SQL database. I first drafted it in Chen notation using the web-based diagramming tool draw.io, which acted as a foundation for the final version constructed in MySQL Workbench, a dedicated SQL database design and management tool. The database defined relationships between Mii (player-created avatar) data on the Nintendo Wii platform, the players who created them, the software they were used in, and the consoles they were stored on. The purpose of this visualisation was to clearly express and communicate these relationships.

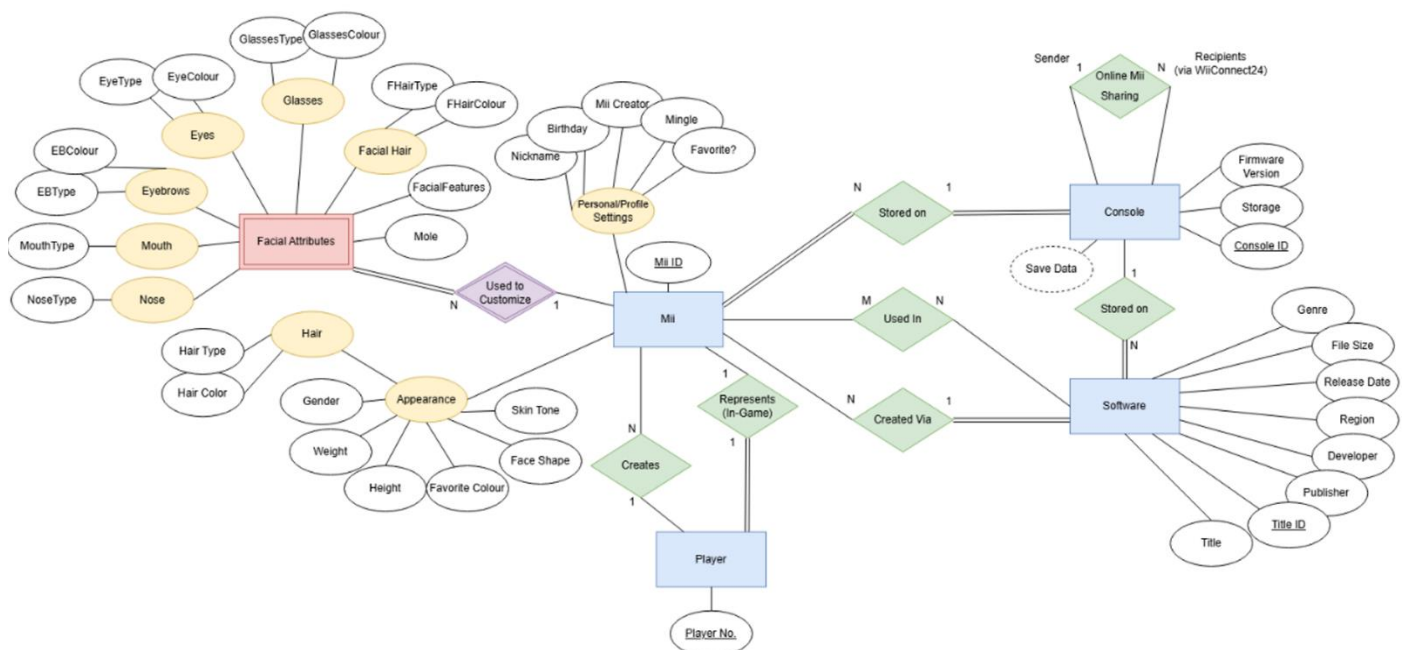


Figure 1a. ER-Diagram (Chen Notation) – draw.io

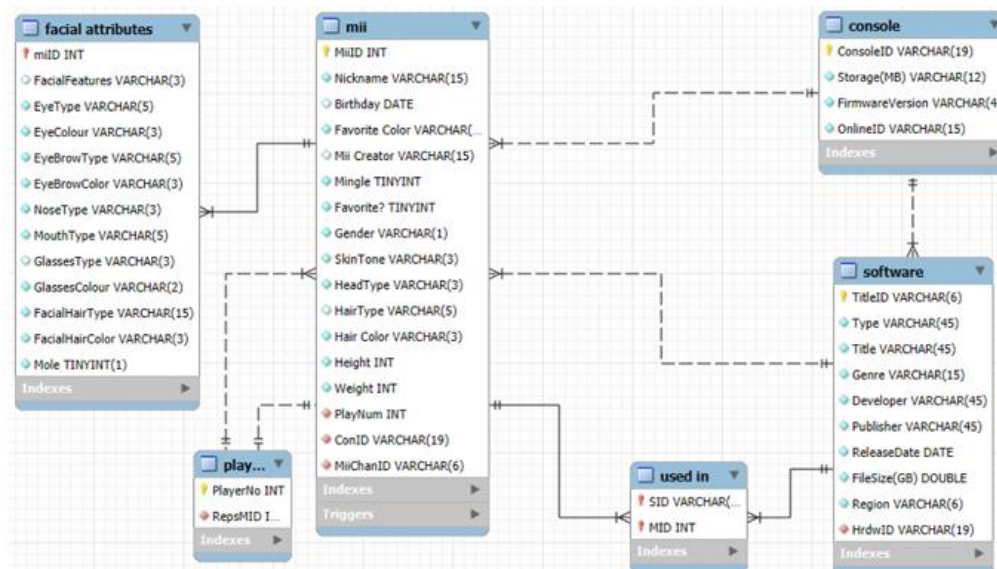


Figure 1b. ER-Diagram – MySQL Workbench

I would consider this visualisation explanatory, as its purpose was to communicate predefined relationships in the database. The ER diagram conveys how the relationships between Mii data, players, software, and consoles were structured and connected, rather than being used to explore or discover new patterns within the database.

Creating the ER diagram proved to be beneficial since it provided an immediate and clear view of how all entities were connected using SQL relationship types (e.g., 1:1, 1:N, M:N). This made it easier to communicate the database structure and document its design in other formats, for example, when mapping the ER diagram to a relational schema as another part of the same assignment.

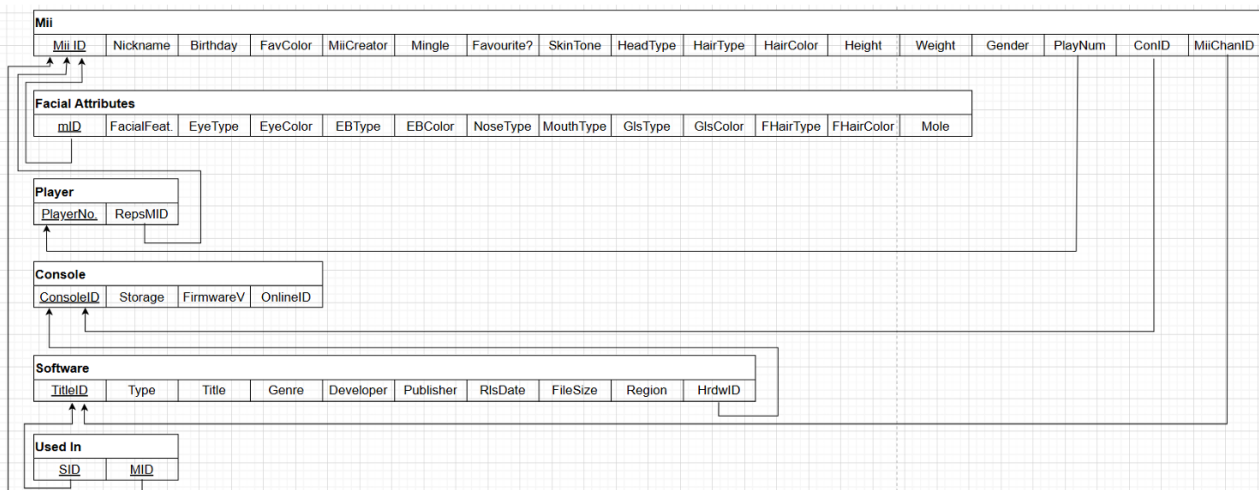


Figure 2. Relational Schema

While the database was represented in other forms in the assignment, such as text descriptions of each entity, raw SQL tables, and the relational schema seen above, using these formats without the visual representation that the ER-diagram provides would make communicating the data less efficient and harder for the reader to interpret.